

```
import cv2
import numpy as np
from google.colab import files
from matplotlib import pyplot as plt
```

```
# Upload image
uploaded = files.upload()
```

cvvv5.png

**cvvv5.png**(image/png) - 290257 bytes, last modified: 7/22/2026 - 100% done  
Saving cvvv5.png to cvvv5.png

```
# Get uploaded file name
image_name = list(uploaded.keys())[0]

# Read the image
img = cv2.imread(image_name)

# Check if image is loaded successfully
if img is None:
    print("Error: Could not load the image.")
else:
    # Convert to grayscale
    gray_image = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Create a 5x5 rectangular kernel
    kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))

    # Perform erosion
    eroded_image = cv2.erode(gray_image, kernel, iterations=1)

    # Save the output image
    cv2.imwrite("eroded_image.jpg", eroded_image)

    # Display images
```

```
plt.figure(figsize=(10,5))

plt.subplot(1,2,1)
plt.imshow(gray_image, cmap='gray')
plt.title("Original Grayscale Image")
plt.axis("off")

plt.subplot(1,2,2)
plt.imshow(eroded_image, cmap='gray')
plt.title("Eroded Image")
plt.axis("off")

plt.show()

print("Eroded image saved as 'eroded_image.jpg'")
```

Original Grayscale Image



Eroded Image



Eroded image saved as 'eroded\_image.jpg'